

Optimized GNSS Network Station Selection to Support the Development of Ionospheric Threat Models for GBAS

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ABSTRACT

Extremely large ionospheric spatial gradients could cause potential integrity threats to the users of Global Navigation Satellite Systems (GNSS) augmentation systems. Thus these ionospheric anomalies need to be monitored by ground reference stations and the users must be alarmed within time-to-alerts. The Long-Term Ionospheric Anomaly Monitoring (LTIAM) tool has been developed to continuously monitor ionospheric behavior over the life cycle of Ground-Based Augmentation Systems (GBAS) and to build ionosphere threat models for all regions where GBAS will be fielded in the future. The use of poor-quality GNSS data degrades the accuracy of ionospheric delay estimates, and thus the performance of the resulting threat models.

Previous work developed a comprehensive method of GPS data quality determination to identify stations that are poor according to multiple quality metrics. This method eliminates the almost 90% of spurious or false gradients when applied to the Continuously Operating Reference Stations (CORS) stations in the Conterminous U.S. (CONUS). However, use of this method end up with eliminating more stations than necessary because this method determined thresholds for each quality parameter independently without considering interrelation between parameters.

This paper develops an optimized method to select well-distributed, high-quality stations for GNSS network. Thresholds which maximize the elimination of spurious gradients while minimizing unnecessary station removal are established. This method achieved 89% reduction on faulty anomaly candidates while discarding only 16% of

the over 1500 CORS stations. Since this method is also used to remove redundant stations in dense regions with a desired baseline constraint, this will benefit to reduce the processing time of LTIAM. These algorithms are also tested using data collected from other GNSS reference station network to verify the performance of proposed method.

1.0. INTRODUCTION

The Long-Term Ionospheric Anomaly Monitoring (LTIAM) tool is an automated software package developed to build ionospheric anomaly threat models from observed events by analyzing past data and continuously monitoring ionospheric behavior over the life cycle of GBAS [1, 2]. The LTIAM tool will be used to verify the current threat model over the life cycle of system and update it if necessary. As the current solar cycle approaches its maximum in 2013-2014, this is of particular importance over the next few years.

The LTIAM software can observe ionospheric gradients over short-baseline distances of 5 – 40 km using data automatically collected from the CORS network, which has over 1500 stations as of 2011 in the CONUS and a few other countries compared to about 400 stations prior to 2004. As the total number of stations increases, the observability of ionospheric behavior improves, but the number of stations with poor GPS data quality also increases. Thus, while automated measurement screening is included in LTIAM, the use of poor-quality data produces many cases of faulty anomaly candidates and adds a great burden to the process.

Previous work developed a comprehensive method of GPS data quality determination to identify stations with poor quality data which produce faulty anomaly candidates. The use of this one dimensional one step search method with CORS stations in CONUS eliminates the almost 90% of spurious or false gradients. However, thresholds to remove poor stations were established independently for each quality metric without considering interrelation between parameters. Because the violation of any one threshold results in exclusion, this method eliminates more stations than necessary.

This paper develops a new station selection algorithm to establish thresholds which maximize the elimination of spurious gradients while minimizing unnecessary station removal. In this algorithm, set of thresholds for the quality parameters is determined by using two dimensional iterative concepts which searches for the most troublesome station and at the same time the most efficient quality parameter. Thresholds are derived for each of these metrics, and stations which exceed the determined threshold are excluded from LTIAM measurement processing unless they are recovered by a secondary check on their location. Stations whose location for observing the ionosphere is sufficiently important are retained despite poor data quality.

Regardless of data quality, a large number of stations which are geographically redundant in dense regions bring an excessive computation load to automated post-processing of data by LTIAM. As long as an area of interest is covered by a sufficient number of stations with separations of less than a desired baseline distance, other nearby stations in the region which do not improve the geographic observability of ionospheric behavior are redundant and can be removed even if they provide high-quality data. This paper therefore proposes a method to select a subset of stations which are geographically important among good-quality stations. Subset of stations which are evenly distributed and maintain an area covered by stations with a desired baseline distance will benefit to reduce the processing time of LTIAM and consequently help to build ionosphere threat models. In this paper, Section 2.0 illustrates the effect of poor quality data; Section 3.0 explains the optimized GNSS network station selection methodology in detail; Section 4.0 and Section 5.0 show the results of applying this method to CORS stations in CONUS and other GNSS reference network respectively, and Section 6.0 concludes the paper.

2.0. POOR CORS DATA QUALITY

This section investigates the effect of GPS data collected from stations with poor data quality on the ionospheric delay and gradient estimation. Figure 1 shows real example of poorly sited CORS stations in CONUS. The data from these stations which are located next to a solar

panel, in a bush, and even in between towers may produce erroneous estimates of ionospheric delays, and consequently unusually large ionospheric gradients due to corrupted raw measurements.



Figure 1. Real Example of Poorly Sited CORS Stations in CONUS

Figure 22 shows the slant ionospheric delays observed from two nearby stations LS02 and VARI while they tracked PRN 15 on 24 May 2012. LS02 is a good example of a station with poor GPS data quality. From the many fragments of ionospheric delay estimates from LS02 (red), it is evident that its carrier-phase measurements are corrupted by numerous cycle slips, resulting in outliers and short arcs of valid measurement and outliers. LS02 (red) also held a small amount of data compared to the normal station VARI (blue). Figure 33 shows the estimated ionospheric spatial gradients between LS02 and VARI. The ionospheric-delay leveling errors due to the short arcs from LS02 are observable at each end of the curve, and the large gradients due to the excessive cycle slips on LS02 are evident in the center of the curve. This example illustrates how poor data quality degrades the accuracy of ionospheric delay estimation and can produce extremely large ionospheric gradients that are not real.

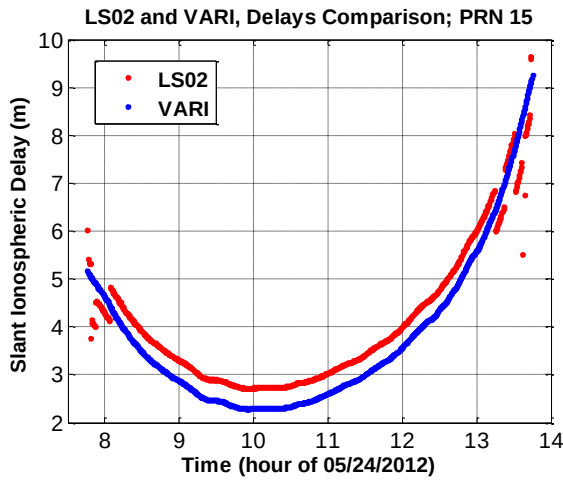


Figure 2. Dual-frequency Slant Ionospheric Delay Estimates for Stations LS02 (Poor Quality Data) and VARI (Good Quality Data)

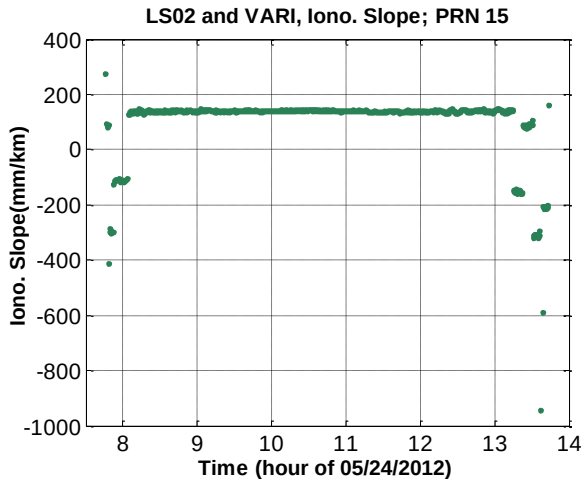


Figure 3. Ionospheric Gradient Estimates Corrupted by Poor Quality Data from Station LS02

3.0. DATA ANALYSIS METHODOLOGY

The methodology for selecting optimized GNSS network stations with well-distributed, high-quality data is composed of three steps: measuring GPS data quality information using several metrics, determining thresholds of each of these metrics to remove stations with poor-quality data, and re-examining not only tentatively excluded stations but also redundant station in a dense area considering their geographical locations. First, data quality information is obtained by processing the RINEX file collected at each station through a series of data-quality-measurement algorithms[.]. Then, stations with sufficiently poor quality data are selected based on whether the quality parameters of the station exceed established thresholds. Third, to observe anomalous ionospheric gradients in all of area, the remained stations should cover all of area with separations of less than 40 km to the degree possible. To meet this criterion in

regions with relatively few stations, some degree of data quality may need to be sacrificed. Therefore, we examine the geographical contribution of stations selected to be removed. Stations whose location increases the geographical observability of ionospheric behavior are restored despite their poor data quality. On the other hand, a good-quality station may be geographically redundant if located in a dense area. To select a subset of stations which are geographically important among good-quality stations, Stations whose location do not help in terms of observability of ionosphere and only increase computational load are removed despite their good data quality. The details of each step are described in the following subsections.

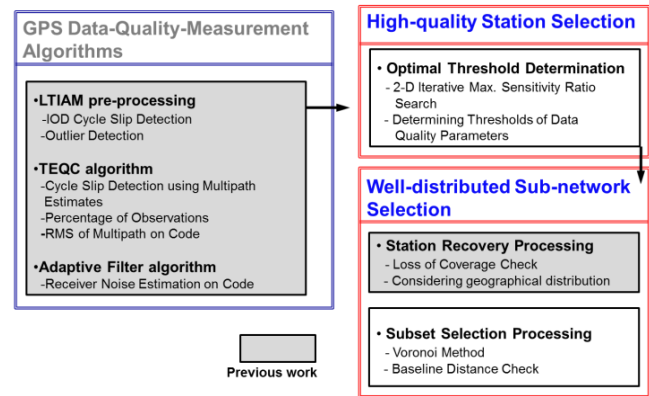


Figure 4. Optimized GNSS Network Station Selection Algorithms

3.1. DATA QUALITY MEASUREMENT ALGORITHMS

The data-quality measurement algorithms were already developed[.]. The input of these algorithms is the RINEX file collected from a station of our interest for two consecutive days and the output is the GPS data quality information of the corresponding station. These algorithms are composed of mainly three parts: LTIAM pre-processing, the “Translation, Editing, and Quality Check (TEQC)” algorithm, and the adaptive filter algorithm. In LTIAM pre-processing step, the number of IOD cycle slips, the number of outliers, and the number of short arcs are counted as separate data-quality measurements. Second, we implemented the TEQC quality metrics which developed by the University Navstar Consortium (UNAVCO), which are commonly used to solve pre-processing problems [.]. The TEQC method is used to obtain quality information, which includes the percentage of observations, the mean of multipath on L1 code and L2 code, and the number of cycle slips detected using multipath estimates. Third, an adaptive filter algorithm is designed to estimate receiver noise on code measurements[.].

3.2. HIGH QUALITY STATION SELECTION

Among the GPS data quality measurement, seven parameters have the greatest impact on LTIAM performance. Those are the number of IOD cycle slips, number of short arcs, number of outliers, percentage of (valid) observations, multipath on L1 code and L2 code, and latency. ‘Latency’ indicates whether the number of days for which the RINEX files are properly loaded. Figure 5 and Figure 6 show the process of determining the stations to be removed based on these seven quality parameters. Once we obtain quality measurements for all stations, we rank stations in order from worst to best for each quality parameter. The worst station is on the top and the best station is in the bottom in Figure 6. In the Figure 6, the black line represents the threshold and the blue box and red box show normal stations and stations generating faulty candidates respectively. As the threshold is reduced, both the number of stations removed, $M_{stations}$, and the number of faulty ionospheric anomaly candidates removed, $N_{candidates}$, increase. We wish to remove as many faulty candidates as possible. However, we also wish to avoid removing too many stations with acceptable data quality in order to remove only a few more faulty candidates. Thus we measure the sensitivity ratio, λ , that expresses the relationship between the number of faulty ionospheric gradients removed and the number of stations removed.

$$\lambda = \frac{N_{candidates}(i)}{M_{stations}(i)};$$

Where i is rank of stations ($i = 1, 2, \dots, \max$)

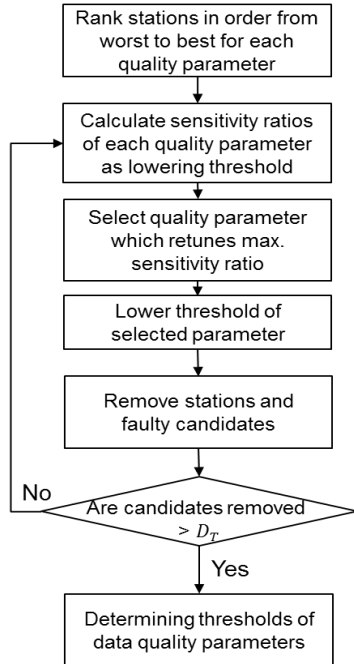


Figure 5. Flow Chart for High-Quality Station Selection Alogrithm

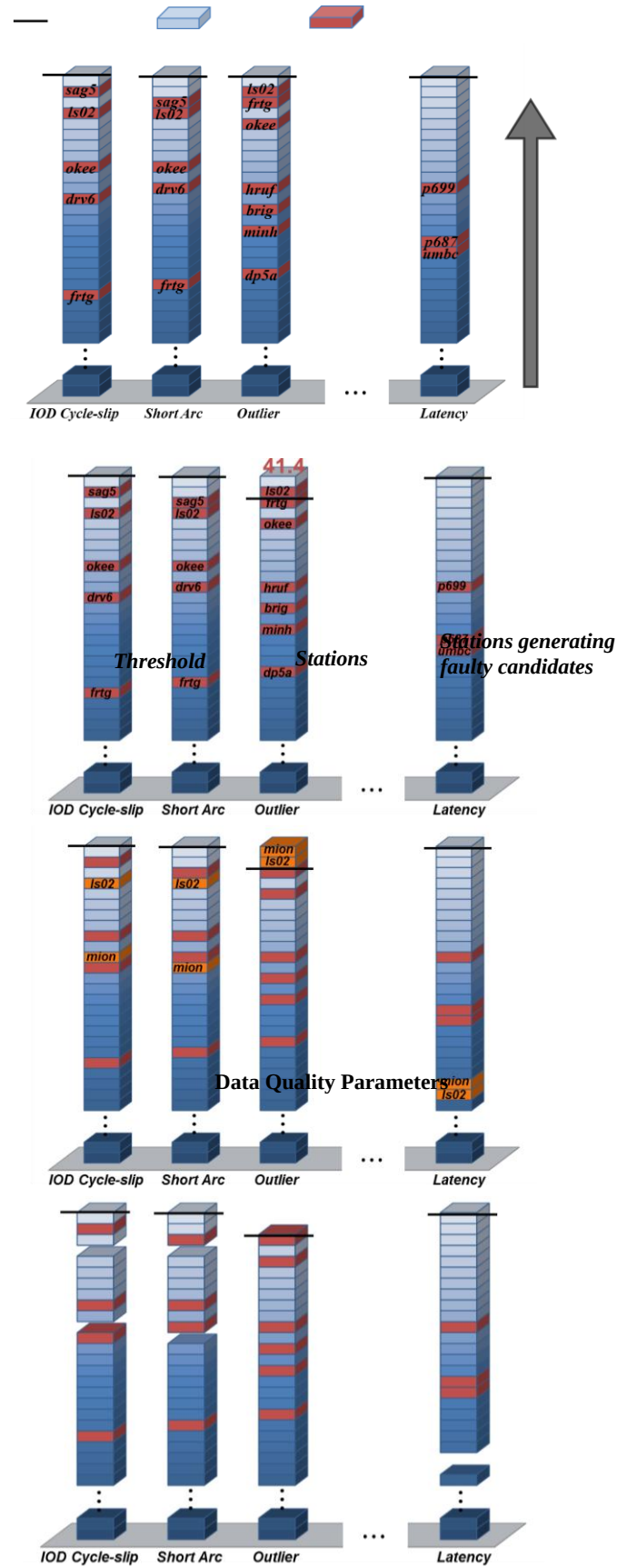


Figure 6

At this point, iteration processing starts. We compute sensitivity ratios of each station as lowering threshold. Next to searches for the most troublesome station and at the same time the most efficient quality parameter, we apply two dimensional concepts. In other words, we compute sensitivity ratios of all quality parameters and select quality parameter which returns the maximum sensitivity ratio among quality parameters. Then we lower the threshold of the selected parameter only in the current step and next remove stations and faulty candidates which exceed the lowered threshold of the selected quality parameter. We iterate this procedure until the desired numbers of faulty anomaly candidates are removed. In other words, the thresholds are determined based on desired percentages of number of faulty candidate removed. 이제까지 설명한 방법이 바로 the two dimensional iterative maximum sensitivity ratio search method 이다. Because the one-step search method which was developed in pervious searches for the troublesome station only, whereas the two dimensional iterative search method searches for the most troublesome station and at the same time the most efficient quality parameter, the current method show better performance. The results of this process in CONUS are shown in Section 4.

3.3. WELL-DISTRIBUTED SUB-NETWORK SELECTION

Stations with poor quality data are determined through the two dimensional iterative search procedure explained in the previous subsection. However, if a particular station significantly increases the geographical observability of ionospheric behavior in an area with relatively few stations, it should be retained despite poor data quality. On the other hand, the stations in dense regions are not significantly improving the geographic observability. Thus, these redundant stations which only bring an excessive computational load should be eliminated even if it has a good quality data. Stations with poor quality data 을 restored 시키는 processing 인 station recovery processing 의 경우, 이전 연구에서 개발 되었다. 이 방법에서는 먼저, for each station, the coverage of other stations within a 100-km radius is examined. "Station coverage" is defined as the area within which pairs of stations whose baseline is less than 100 km form. If a poor-quality station to be removed has another station nearby, the change in station coverage will be small, even after the poor-quality station is removed. However, if not many nearby stations exist, the coverage loss after discarding a station will be lager. This is loss of coverage of a station. As a rule of thumb, if the loss of coverage after discarding a station is more than 30% of the original coverage, that station is restored despite its poor data

quality. the geographical observability 을 고려하며 Redundant stations in dense regions 을 제거 하는 것이 the final step of optimized GNSS network selection 이다. This subset selection processing includes two methods. One is used to make subset of stations to be evenly distributed and the other is used to maintain an area covered by stations with a desired baseline distance. The first method is Voronoi method which is the commonly used method to divide space into a number of regions[]. The station in dense region has a smaller Voronoi region and thus to remove redundant stations we remove stations with the smallest Voronoi region first as shown in the Figure 7 as dark gray area. After a station is discarded, we redistributed the Voronoi diagram of all of remaining stations and iterate processing which throw away a station with smallest Voronoi region. However the Voronoi method is not considering the loss of coverage. Thus before discarding the station, we perform the baseline distance check. 이 방법에서 we check coverage by stations with separations of less than a desired baseline constraint. For example, if a desired baseline constraint was 40km as shown Figure 8, for a station, the coverage of other stations within a 40-km radius is examined. Thus station coverage is defined as the area within which pairs of stations whose baseline is less than 40 km form. When the loss of coverage is less than 10%, we discard the station which is chosen by the Voronoi method.

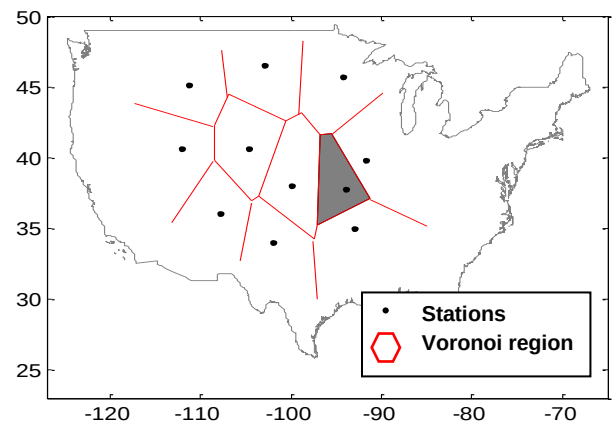


Figure 7. Overview of Stations with Voronoi Region (Red Line)

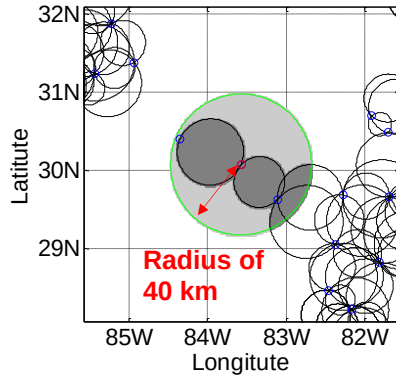


Figure 8. Check coverage by stations with separations of less than 40 km

4.0. RESULTS OF CORS STATION SELECTION

The dates from which CONUS data were collected and analyzed to evaluate the performance of the optimized GNSS network selection algorithms are shown in 오류! 참조 원본을 찾을 수 없습니다.. The geomagnetic conditions on these seven consecutive days are shown with two indices of global geomagnetic activity from space weather databases: planetary K (Kp) and disturbance storm time (Dst). In this period, a total of 1587 CORS network stations were operating in CONUS.

Table 1. Dates Analyzed to Investigate CORS Network Station Quality in CONUS

Day (UT mm/dd/yy)	Kp	Dst
24/05/12	2.0	-15
25/05/12	2.3	17
26/05/12	2.3	-6
27/05/12	1.3	14
28/05/12	2.3	23
29/05/12	2.3	23
30/05/12	2.3	16

As Kp and Dst in 오류! 참조 원본을 찾을 수 없습니다. indicate, the days on which the geomagnetic storm condition was very quiet was chosen. This allows CORS station data quality to be observed while minimizing any influence of abnormal ionospheric behavior. Since it is known that an anomalous ionospheric event did not occur in this period, the threat candidates that result from LTIAM processing are known to be faulty candidates generated by processing of poor-quality data.

The results from the GPS data-quality measurement algorithms for station and the rank (in order of bad quality) for each quality parameter are illustrated in []. 이 results 중 관심이 있는 seven parameters 를 이용하여

high-quality station selection process 를 진행하게 되고, 그 결과는 Figure 9 and 10 보여지고 있다.

As shown in Figure 9, once data from CORS stations in CONUS for seven (or longer) consecutive days are collected and then we obtain statistical distributions of each quality parameter. Data points exceeding $\mu + 9\sigma$ (the mean plus nine times the sample standard deviation) are classified as extreme outliers. After discarding these outliers, we obtain a nominal distribution for each. Using this revised distribution, the threshold is represent as mean plus k sigma of each parameter. Figure 10 shows the k values of seven quality parameters at each iteration step. As the steps are preceded, the threshold and k values decrease.

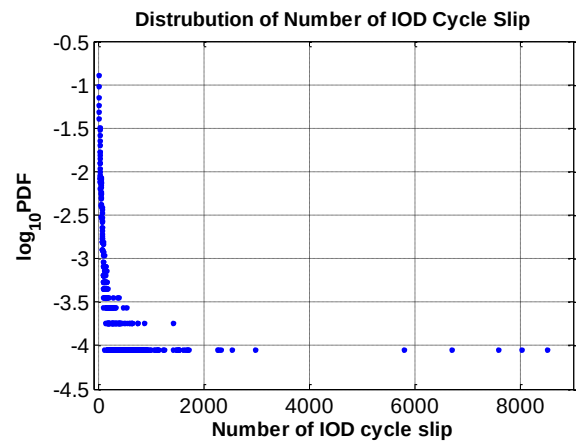


Figure 9. Probability density function of IOD cycle slip for each station per day (data collected for 7 days)

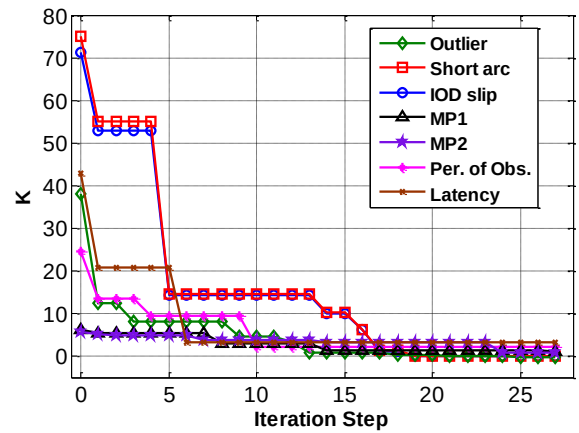


Figure 10. K Values of All Seven Quality Parameters

Accordingly the stations are discarded more and more, Figure 10 shows the percentage of the number of faulty candidates removed as the red curve and the percentage of the number of stations removed as the blue curve at each iteration step. The other curves represent the percentage of the number of stations which exceed the threshold of

the corresponding parameter. Since the number of outliers has the strongest influence on producing faulty candidates, the percentage of the number of stations which exceed the threshold of outlier which is shown as green is the closest to the blue curve. LTIAM processing of all stations on seven consecutive days generated 484 faulty candidates (non-existent ionospheric anomalies). As illustrated in Figure 10, 100% of faulty candidates are removed at 27th iteration, but with a price of removing 60% of stations. If the desired percentage of number of faulty candidates removed was chosen as 90%, it removes only 20% of the stations as shown at 14th iteration. 이때의 threshold of

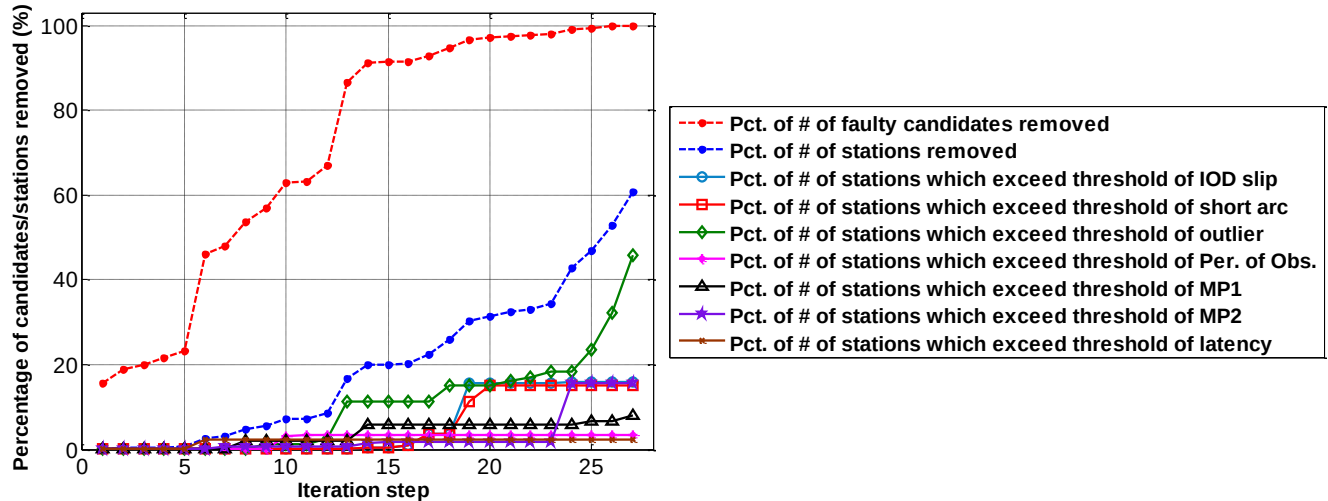


Figure 11. Percentage of number of stations/candidates removed at each iteration step

Table 2. Results of High Quality Station Selection on CORS Network in CONUS (based on LTIAM results from 24-30 May 2012)

	# of candidate removed on seven days (Out of 484)	# of stations removed (Out of 1587)
2D iterative search method (Current)	441 (91.1%)	316 (19.9%)
1D one-step search method (Previous)	420 (86.8%)	323 (20.4%)

We performed the station recovery processing on 316 stations which were classified as bad stations initially. Among those stations, 57 stations were deemed to be "geographically critical" stations and were not removed. These 57 stations and stations removed with poor quality data are shown in green and red respectively in Figure 11.

each quality parameters 는 Table 2 에 나타나 있다. Table 2 shows that this result is certainly better than that got previously from one dimension one-step search method. the current method 는 the previous method 와는 달리 the most troublesome station 과 the most efficient quality parameter 를 같이 고려하여 threshold 를 정하기 때문에, the two dimensional iterative search method removes more faulty candidates, whereas the stations removed are less.

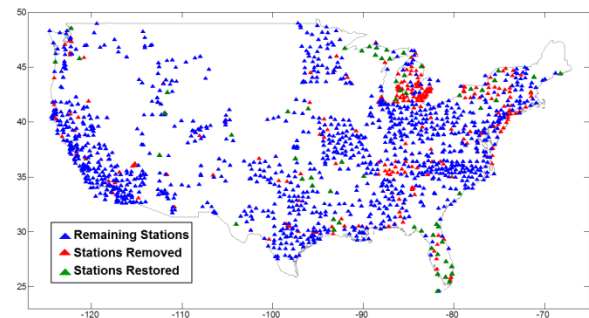


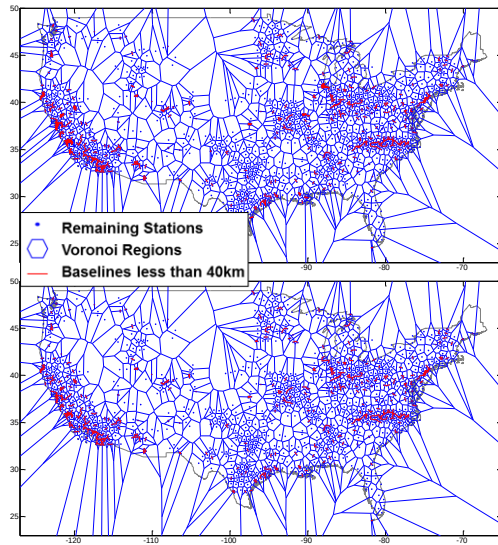
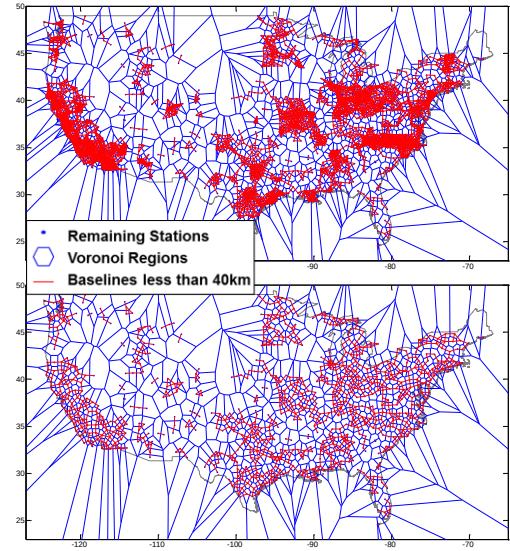
Figure 12. Map of CORS Stations in CONUS (Stations Removed in Red; Stations Restored in Green)

오류! 참조 원본을 찾을 수 없습니다.3 summarizes the results from the station recovery processing. After restoring these 57 stations, the number of faulty ionospheric anomaly candidates removed don't change much compared to the results before the station recovery processing. The high quality station selection and the station recovery processing removed 89% of the total false anomalies while discarding only about 16% of the total stations.

Table 3

	After high-quality station selection	After station recovery
# of stations removed in CONUS (Out of 1587)	316 (19.9%)	259 (16.3%)
# of faulty ionospheric anomaly candidates removed on seven days (Out of 484)	441 (91.1%)	432 (89.3%)

Figure 12 and 13 show the Voronoi diagram of the stations which were survived from the high quality station selection and the station recovery processing. 두 figure 는 각각 subset selection 의 baseline constraint 가 40km 일 때와 100km 일 때의 결과를 보여주고 있다. Because the number of station pairs with baseline less than 40km is relatively small, only about 5% of the stations in a dense area are removed by the subset selection processing as shown in Figure 12. However, as illustrated in Figure 13 and Table 4, if baseline constraint was set 100 km, the subset selection reduced stations down to 46% of the total stations. This final result shows that the stations are evenly distributed and the coverage with the desired baseline distance is well maintained.

**Figure 13****Figure 14****Table 4**

	After station recovery processing	After subset selection processing
# of stations with a Baseline Constraint of 40km (Out of 1587)	1328 (83.7%)	1242 (78.3%)
# of stations with a Baseline Constraint of 100km (Out of 1587)	1328 (83.7%)	734 (46.3%)

5.0. RESULTS OF OTHER GNSS NETWORKS STATION SELECTION

To validate the performance of the high-quality, well-distributed station selection method, the proposed tool was also tested for other GNSS networks, including Japanese, European, and Canadian GNSS networks. Japanese GPS network 로는 GEONET data 를 이용하였으며, European GPS network 로는 EUREF, GREF, IGS, FReDNet 을 Canadian GPS network 로는 CACS, CHAIN 의 데이터를 이용하였다. The dates from which data were collected and analyzed to test the performance of the optimized GNSS network selection algorithms 는 CORS 의 데이터를 처리했을 때와 같은 날이다.

In this period, a total of 1229 GEONET network stations were operating in Japan. Among those total stations, 238 (19.4%) of all stations were chosen to be removed due to poor data quality after high quality station selection process. If the 238 stations selected from this quality check were removed, 479 (90.4%) of all faulty candidates

out of a total of 530 faulty candidates would disappear, as shown in 오류! 참조 원본을 찾을 수 없습니다.5. These results are similar with the case of CORS network. This indicates that the qualities of two networks are also similar.

Table 5

	After high-quality station selection	After station recovery
# of stations removed in GEONET (Out of 1229)	238 (19.4%)	236 (19.2%)
# of faulty ionospheric anomaly candidates removed on seven days (Out of 530)	479 (90.4%)	477 (90.0%)

Figure 14 shows result of station recovery process for GOENET. Since the GEONET forms a dense network in compere to the CORS network, only two stations were considered to be "geographically critical" stations among the 238 stations with poor quality data. 또한 GEONET 의 경우 many pairs of stations with baselines less than 40 km 가지고 있음으로, Figure 15 과 table 6 에 나타나 있듯이, subset selection process with a baseline constraint of 40 km 을 통해서 stations 을 약 전체의 40%까지 줄일 수 있다. 즉, we need to process only about 40% of GEONET stations and yet maintain the similar level of ionospheric obervability.

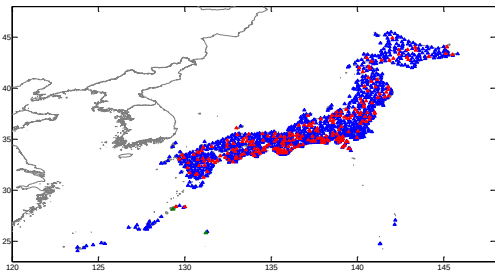


Figure 15

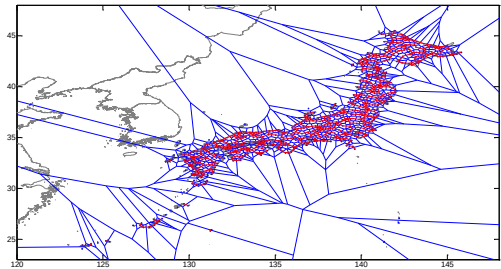
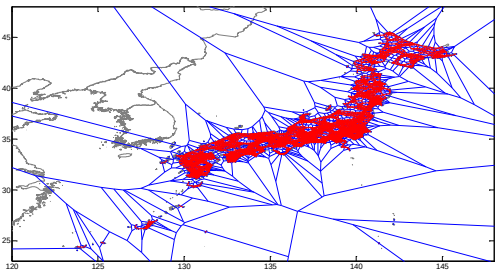


Figure 16

Table 6

	After station recovery processing	After subset selection processing
# of stations with a Baseline Constraint of 40km (Out of 1229)	993 (80.8%)	473 (38.5%)

Results from European and Canadian GPS networks 가 Figure16 과 17 에 각각 보여지고 있다. 이 두 network 의 경우, station 의 개수가 너무 많지 않음으로 station recovery processing 까지만 test 를 진행 하였다. Table 7 을 보면, European GPS networks 의 경우, 총 246 개의 station 중 29(12%)의 station 이 제거 되며 이를 통해 약 94%의 faulty candidates 가 제거된다. 또한 Canadian GPS networks 의 경우 45 개의 station 중 only 7(16%)의 station 이 제거됨으로써 100% 모든 faulty ionospheric anomaly candidates 가 제거되는 것이 table 8 에 나타나 있다.

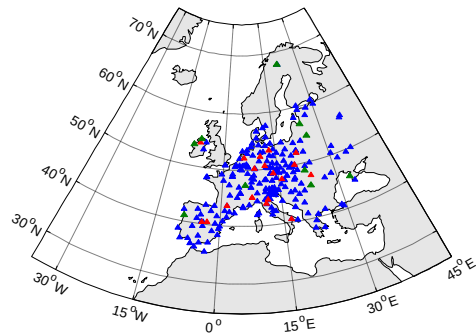


Figure 17

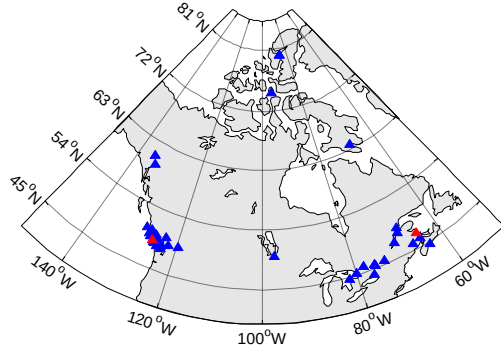


Figure 18

Table 7

	After station recovery
# of stations removed in European (Out of 246)	29 (11.8%)
# of faulty ionospheric anomaly candidates removed on seven days (Out of 32)	30 (93.8%)

Table 8

	After station recovery
# of stations removed in Canadian (Out of 45)	7 (15.6%)
# of faulty ionospheric anomaly candidates removed on seven days (Out of 48)	48 (100%)

Other GNSS network 의 결과가 CORS network 의 결과보다 같거나 더 낮은 성능을 보여주고 있으므로 이 논문에서 제안한 optimized GNSS network selection method 이 좋은 성능을 나타내는 것을 알 수 있다.

6.0. SUMMARY

This paper presents the results of the optimized method to select GNSS network stations with well-distributed, high-quality data and verify the performance of proposed method. Poor quality data 를 갖는 특정한 station 은 LTIAM 에 원치 않는 많은 faulty candidates 를 발생시키며, 이는 cause excessive processing time in the manual validation. The optimized method which was developed in this paper achieved 89% reduction on faulty anomaly candidates while discarding only 16% of CORS stations. Results from processing recently-collected GNSS data confirm the proposed methods perform well

also for other GNSS network. Geographical observability of ionosphere with a baseline constraint of 40km can be maintained by processing only 40% of GEONET stations. Consequently, 이 방법은 faulty candidates 로 인해 manual validation 과정에서 발생하는 excessive processing time 과 노력을 greatly 줄이며, redundant station 의 데이터를 처리함으로써 발생하는 불필요한 processing time 단축시킴으로, this should benefit future to build ionosphere threat models for all regions where GBAS will be fielded in the future.